

Images can be very easily displayed and manipulated inside a canvas. You can stamp or stretch an image, use transformations on it, and even make your image the focus of the entire canvas. The HTML5 Canvas API includes a few simple commands for adding image content to the canvas.

However keep in mind that images can also add a complication to the canvas operations as we have to wait for them to load. Browsers will usually be loading images asynchronously as your page script is rendering. However, if you attempt to render an image onto a canvas before it has completely loaded, the canvas will fail to render any image at all. As such, you should be careful to make sure the image is loaded completely before you attempt to render it.

The canvas drawing context defines a `drawImage()` method for drawing an image on a canvas. The method can take three, five, or nine arguments.

- ▶ `drawImage(image, dx, dy)` takes an image and draws it on the canvas. The given coordinates (dx, dy) will be the upper-left corner of the image. Coordinates (0, 0) would draw the image at the upper-left corner of the canvas.
- ▶ `drawImage(image, dx, dy, dw, dh)` takes an image, scales it to a width of dw and a height of dh, and draws it on the canvas at coordinates (dx, dy).
- ▶ `drawImage(image, sx, sy, sw, sh, dx, dy, dw, dh)` takes an image, clips it to the rectangle (sx, sy, sw, sh), scales it to dimensions (dw, dh), and draws it on the canvas at coordinates (dx, dy).

As we can see, in order to draw an image on a canvas, first of all we need an image. The image can be an existing `<img>` element, or you can create an `Image()` object with JavaScript. Either way, you need to ensure that the image is fully loaded before you can draw it on the canvas.

If you're using an existing `<img>` element, you can safely draw it on the canvas during the `window.onload` event. If you're creating the image object entirely in JavaScript, you can draw the image on the canvas during the `Image.onload` event.

Using Gradients

Gradients can be applied in the form of a gradual algorithmic sampling of colors. You can employ two different styles to do the same - stroke style or fill style. Creating gradients is a three-step process:

1. Create the gradient object itself.
2. Apply color stops to the gradient object, signaling changes in color along